

TECHBIG SOLUTIONS

COMPLETE VISUAL STUDIO & .NET DEVELOPMENT SYLLABUS

Location: Ambattur, Chennai

Training Focus: Visual Studio • C# • .NET • ASP.NET Core • Web API • SQL Server • EF Core • Git • Azure • Deployment

Course Objective

This comprehensive program is designed to take learners from Visual Studio and C# fundamentals through professional .NET application development. It combines desktop, web, REST API, database, security, testing, DevOps, cloud deployment and hands-on projects. The syllabus is aligned with the current Microsoft .NET/ASP.NET Core learning ecosystem, including modern Visual Studio workflows.

Item	Recommended Coverage
Level	Beginner to Advanced
Suggested Duration	3–6 months, depending on pace and project depth
Mode	Classroom / Online / Practical Lab
Prerequisites	Basic computer knowledge; programming basics helpful but not mandatory
Primary Tools	Visual Studio, .NET SDK, C#, SQL Server/LocalDB, Git
Outcome	Build, test, debug, deploy and maintain production-style .NET applications

1. Visual Studio & .NET Development Environment

- Visual Studio overview, editions and project templates
- Visual Studio Installer, workloads, components and SDK selection
- Solution Explorer, Properties, Toolbox, Output and Error List
- Projects, solutions, configurations and build profiles
- NuGet Package Manager and package/version management
- .NET SDK, runtime, target frameworks and project files
- Command-line development with dotnet CLI
- Git integration, repositories and basic source control workflow

2. C# Programming Fundamentals

- Variables, constants, data types, type conversion and operators
- Input/output, strings, arrays and collections
- Conditional statements and loops
- Methods, parameters, return values and optional parameters
- Classes, objects, constructors, properties and access modifiers
- Inheritance, polymorphism, abstraction and interfaces
- Generics, delegates, events and lambda expressions
- Exception handling and custom exceptions
- LINQ fundamentals and common query patterns
- Asynchronous programming with async/await and Task

3. Windows Desktop Application Development

- Windows Forms project creation and application structure
- Controls, forms, menus, dialogs and layouts
- Events and event-driven programming
- Data validation and user input handling
- File handling and configuration
- Data binding and reusable components
- Debugging desktop applications
- WPF introduction, XAML, controls, layouts, styles and data binding
- MVVM concepts and command binding

4. ASP.NET Core Web Development

- ASP.NET Core architecture and application lifecycle
- Project templates and web application structure
- HTTP, routing, middleware and request pipeline
- Dependency Injection and service lifetimes
- Configuration, appsettings and environment-specific settings
- Razor Pages and Razor syntax
- MVC architecture: Models, Views and Controllers
- Tag Helpers, model binding and validation
- Session, cookies, caching and application state
- Logging and error handling

5. Web API & REST Services

- REST principles and HTTP methods
- ASP.NET Core Web API project structure
- Controllers, endpoints and routing
- CRUD operations and status codes
- DTOs, model binding and validation
- JSON serialization and deserialization
- Swagger/OpenAPI and API documentation
- Authentication and authorization concepts

- API versioning, CORS and security fundamentals
- Consuming APIs from applications

6. Database Programming & Entity Framework Core

- SQL Server and LocalDB integration
- Connection strings and database configuration
- ADO.NET fundamentals
- Entity Framework Core architecture
- DbContext, entities and relationships
- Code First and migrations
- CRUD with EF Core
- LINQ-to-Entities queries
- One-to-one, one-to-many and many-to-many relationships
- Transactions, concurrency and performance basics
- Database seeding and deployment considerations

7. Front-End Integration

- HTML5 and semantic markup
- CSS3 fundamentals and responsive design
- JavaScript fundamentals for .NET developers
- DOM, events and form validation
- Bootstrap and responsive UI
- AJAX/fetch and API integration
- Client-side and server-side validation
- Accessibility and browser developer tools

8. Advanced .NET Development

- Generics, reflection and attributes
- Dependency Injection design patterns
- Repository and service-layer patterns
- Clean Architecture concepts
- Configuration and Options pattern
- Background services and hosted services
- Caching and performance optimization
- File upload/download and streaming
- JSON, XML and serialization
- Parallel programming and asynchronous patterns

9. Security

- Authentication vs authorization
- ASP.NET Core Identity fundamentals
- Role- and policy-based authorization
- Password and credential security
- JWT bearer authentication
- OAuth/OpenID Connect concepts
- HTTPS and secure configuration
- Input validation and protection against common web vulnerabilities
- Secrets management and environment variables

10. Testing & Debugging

- Visual Studio debugging workflow
- Breakpoints, watches, call stack and immediate window
- Debug vs Release builds

- Unit testing with .NET test frameworks
- Test projects, assertions and test organization
- Mocking concepts and dependency isolation
- Integration and API testing
- Logging and diagnostics
- Performance profiling and troubleshooting

11. Git, CI/CD & DevOps

- Git fundamentals and Visual Studio Git tools
- Branches, commits, merges and pull requests
- Conflict resolution
- Build automation
- Continuous Integration and Continuous Delivery concepts
- GitHub/Azure DevOps integration
- Automated testing in pipelines
- Application packaging and deployment

12. Cloud & Deployment

- Publishing .NET applications
- Framework-dependent vs self-contained deployment
- Windows/IIS hosting basics
- Azure App Service deployment concepts
- Application configuration in cloud environments
- Environment variables and connection strings
- Containers and Docker fundamentals
- Monitoring, logging and application health

13. Modern .NET & Cross-Platform Development

- Modern .NET architecture and cross-platform development
- Console, class library and web project types
- .NET MAUI overview for cross-platform applications
- Blazor overview and component-based UI
- Minimal APIs
- Modern C# language features
- Cloud-native .NET concepts

14. Practical Projects

- Project 1: Student Management System with SQL Server
- Project 2: Employee/HR Management Web Application
- Project 3: Inventory Management System
- Project 4: RESTful Web API with authentication
- Project 5: E-commerce mini application
- Project 6: Helpdesk/Ticket Management System
- Project 7: Publish an ASP.NET Core application to IIS or Azure
- Project 8: Capstone project with Git, testing, database and deployment

15. Interview & Career Preparation

- C# and .NET technical interview questions
- ASP.NET Core and Web API interview preparation
- SQL and Entity Framework interview questions
- Debugging and real-world problem-solving exercises
- Git and CI/CD interview preparation
- Project explanation and technical documentation

- Resume/project portfolio guidance
- Mock interviews and practical coding assessments

Recommended Capstone Deliverables

Source code repository • Database schema • API documentation • Test cases • Deployment guide • User/admin documentation • Final project presentation

Note: Microsoft's current documentation covers C#, .NET, ASP.NET Core, Web API, Entity Framework Core, Visual Studio and cloud deployment as core parts of the modern .NET development ecosystem. This syllabus is structured around those areas. ■cite■turn0search1■turn0search3■turn0search2■

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